ASSIGNMENT 6 — QUESTION 7

Q7 - The Lazy Loader: Promise Combinator Practice

JavaScript (ES6+) solution with comments

```
// Q7: Promise.allSettled() reports every result, including failures.

function loadModule(name, milliseconds) {
    return new Promise((resolve, reject) => {
        setTimeout(() => {
            // Randomly reject a module (about 30% chance).
            if (Math.random() < 0.3) {
                reject(new Error(name + " failed to load"));
            } else {
                resolve(name + " Loaded");
            }
        }, milliseconds);
    });
}

async function loadDashboard() {
    const startTime = Date.now();

    const results = await Promise.allSettled([
        loadModule("Profile", 2000),
        loadModule("Posts", 1500),
        loadModule("Messages", 1000)
    ]);

    results.forEach((result, index) => {
        const moduleNames = ["Profile", "Posts", "Messages"];

        if (result.status === "fulfilled") {
            console.log(moduleNames[index] + " succeeded:", result.value);
        } else {
            console.error(moduleNames[index] + " failed:", result.reason.message);
        }
    });

    const totalTime = Date.now() - startTime;
    console.log("Total time taken:", totalTime, "ms");
}

loadDashboard().catch(error => console.error("Unexpected error:", error));
```